

Name _____

Period _____ Date _____

Practice Test – Advanced Kinematics (Version 2)

On any problems, you must show all work, include units, round properly, and box your final answer.

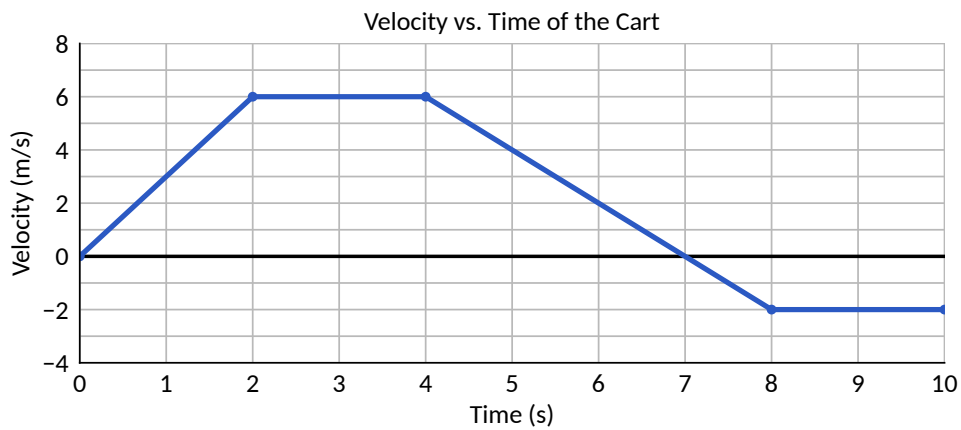
Do the algebra first, before plugging in values.

Rewrite the following statement in the space below: On my honor, before God, I have neither given, nor received, any aid on this test.

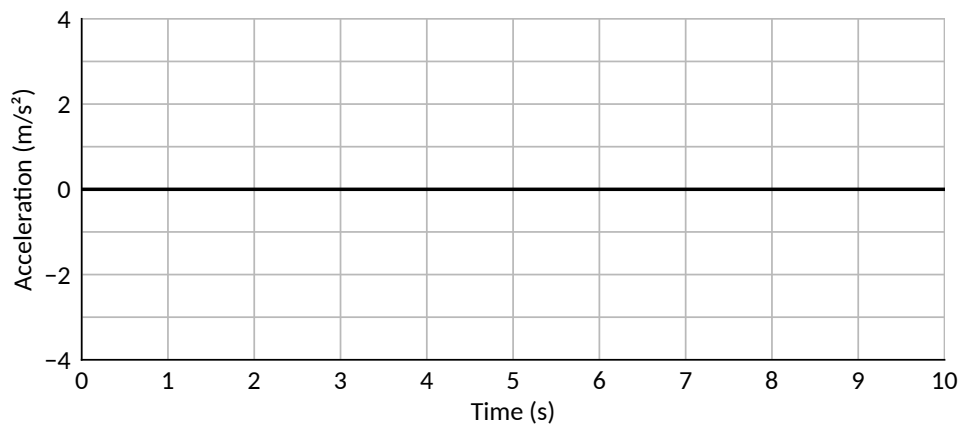
For all questions:

- Use $g = 9.80 \text{ m/s}^2$ and neglect air resistance.
- Round numerical answers to three significant figures.
- 50 minutes. Four questions, 25 points each (100 points total).
- The last page is optional extra practice and is not part of the timed test.

1. (25 points) The velocity-time graph below shows the motion of a cart moving along a straight track from $t = 0$ s to $t = 10$ s. On the blank grid, draw the corresponding acceleration-time graph. Show the slope calculation for each segment of the velocity-time graph.



Draw your acceleration-time graph here:



Slope calculations:

2. (25 points) A car travels along a straight, level road at a constant 27.0 m/s. The driver then brakes, and the car slows with a constant acceleration of magnitude 4.50 m/s^2 until it stops. How far does the car travel while braking?

3. (25 points) A football is kicked from ground level on a level field with an initial speed of 22.0 m/s at 40.0° above the horizontal. It lands at the same height from which it was kicked. Air resistance is negligible. What is the ball's range (the horizontal distance it travels before landing)?

4. (25 points) The roof of a skyscraper is a height H above the street. A small object is released from rest at the roof and falls straight down to the street. Air resistance is negligible. Derive an expression for the velocity of the object just before it hits the street. Express your answer in terms of H and physical constants, as appropriate, and indicate the direction of the velocity.

Optional extra practice (not part of the timed test, not scored)

E1. For the football in Question 3: (a) What is its hang time? (b) What is its maximum height?

E2. For the car in Question 2: (a) How long does it take to stop? (b) If its initial speed were doubled and it braked with the same acceleration, by what factor would its braking distance change?

E3. For the object in Question 4: write its velocity $v_y(t)$ as a function of the time t since it was released, and state the time interval over which this expression is valid.

Answer Key – Practice Test, Version 2 – Advanced Kinematics

Worked step by step, the same way as the class notes: sketch, list the variables, pick the equation, algebra first, then numbers with units. $g = 9.80 \text{ m/s}^2$; answers to three significant figures.

AP Physics 1 equation sheet, kinematics (Mechanics and Fluids, first three):

#1 $v_x = v_{x0} + a_x t$ · **#2** $x = x_0 + v_{x0}t + \frac{1}{2}a_x t^2$ · **#3** $v_x^2 = v_{x0}^2 + 2a_x(x - x_0)$

Same equations as $v_f = v_i + at$, $\Delta x = v_i t + \frac{1}{2}at^2$, $v_f^2 = v_i^2 + 2a\Delta x$. Pick the one that leaves out the variable you don't know and don't need. For vertical motion use y in place of x .

1. Acceleration-time graph (25 points)

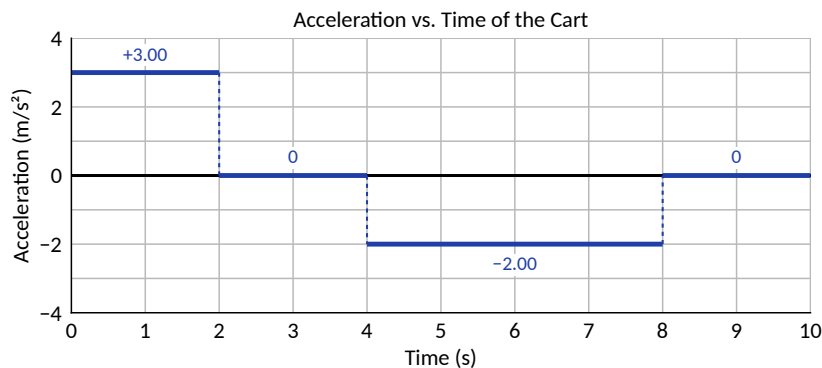
Step 1 Acceleration is the slope of the v-t graph.

$$a = \frac{\Delta v}{\Delta t} = \frac{v_f - v_i}{t_f - t_i} \text{ (read } v_i \text{ and } v_f \text{ at the start and end of each straight piece; values read off grid lines are treated as exact)}$$

Step 2 Find the slope of each segment.

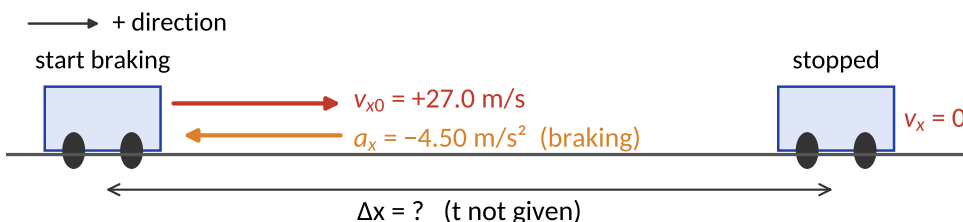
segment	$\Delta v = v_f - v_i$	$\Delta t = t_f - t_i$	$a = \Delta v / \Delta t$
0 s to 2 s	6 m/s - 0 m/s = 6 m/s	2 s - 0 s = 2 s	+3.00 m/s ²
2 s to 4 s	6 m/s - 6 m/s = 0 m/s	4 s - 2 s = 2 s	0 m/s ²
4 s to 8 s	(-2 m/s) - 6 m/s = -8 m/s	8 s - 4 s = 4 s	-2.00 m/s ²
8 s to 10 s	(-2 m/s) - (-2 m/s) = 0 m/s	10 s - 8 s = 2 s	0 m/s ²

Step 3 Draw each slope as a flat line over its time interval. Label both axes with units.



Watch out: the cart turns around where v crosses 0, but the slope does not change there, so a does not change either. A straight slanted v-t line always becomes a flat a-t line; a flat v-t line becomes $a = 0$.

2. Braking car: stopping distance (25 points)



Step 1 Pick a positive direction and get the sign of a.

Let the direction the car moves be +. The car is moving + but slowing down, so its velocity is decreasing: Δv is negative, which makes a negative. Braking acceleration points opposite the motion, so $a = -4.50 \text{ m/s}^2$.

Step 2 List the five variables.

$v_i (= v_{x0})$	$v_f (= v_x)$	a	t	Δx
+27.0 m/s	0 m/s (stops)	-4.50 m/s ²	not given, not asked	? (wanted)

Step 3 Pick the equation.

AP sheet	equation	leaves out
#1	$v_x = v_{x0} + a_x t$	$x - x_0$ (Δx)
#2	$x = x_0 + v_{x0} t + \frac{1}{2} a_x t^2$	v_x (final velocity)
#3	$v_x^2 = v_{x0}^2 + 2a_x(x - x_0)$	t (time) ← use this one

Why #3: time is not given and not asked for. #3, written $v_f^2 = v_i^2 + 2a\Delta x$, is the only one without t .

Step 4 Start from $v_f^2 = v_i^2 + 2a\Delta x$ and set $v_f = 0$. Solve for Δx before any numbers.

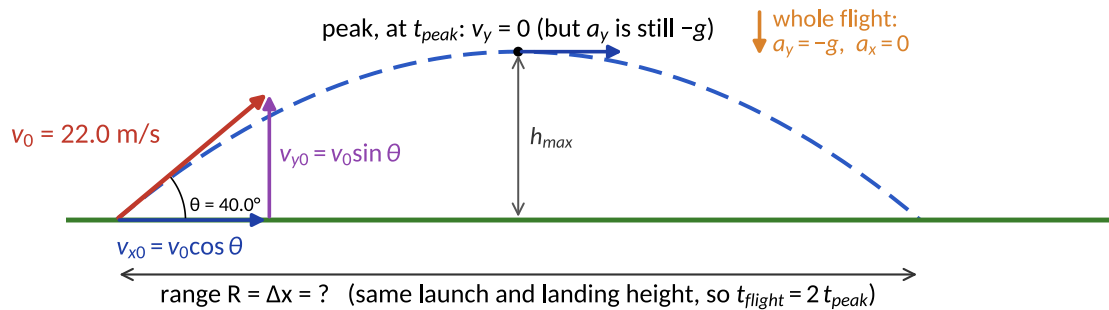
$$v_f^2 = v_i^2 + 2a\Delta x \rightarrow 0 = v_i^2 + 2a\Delta x \rightarrow 2a\Delta x = -v_i^2 \rightarrow \Delta x = -\frac{v_i^2}{2a}$$

Step 5 Plug in numbers with units.

$$\Delta x = -\frac{(+27.0 \text{ m/s})^2}{2(-4.50 \text{ m/s}^2)} = -\frac{(729 \text{ m}^2/\text{s}^2)}{(-9.00 \text{ m/s}^2)} = \boxed{81.0 \text{ m}}$$

The minus sign in the formula and the negative a cancel, so Δx comes out positive: the car moves forward while it stops. If you get a negative distance, a had the wrong sign. Units: $(\text{m}^2/\text{s}^2) \div (\text{m/s}^2) = \text{m}$.

3. Football: range (25 points)



Split the kick into two motions. Sideways moves at a constant speed ($a_x = 0$). Up and down is free fall ($a_y = -g$). **Time** is the only thing both share.

Step 1 Resolve the initial velocity into components (keep letters; numbers go in at the end).

$v_{x0} = v_0 \cos \theta$ (the side next to the angle: along the ground)

$v_{y0} = v_0 \sin \theta$ (the side across from the angle: straight up)

Step 2 Make the T-chart.

horizontal	vertical
$a_x = 0$	$a_y = -g = -9.80 \text{ m/s}^2$
$v_{x0} = v_0 \cos \theta$	$v_{y0} = v_0 \sin \theta$
$\Delta x = R = ?$	at the peak: $v_y = 0$
$t = t_{\text{flight}}$	$t_{\text{peak}} = ?$

Step 3 Find the time to the peak, where $v_y = 0$.

AP sheet	equation	leaves out
#1	$v_y = v_{y0} + a_y t$	$y - y_0 (\Delta y) \leftarrow \text{use this one}$
#2	$y = y_0 + v_{y0} t + \frac{1}{2} a_y t^2$	v_y (final velocity)
#3	$v_y^2 = v_{y0}^2 + 2 a_y (y - y_0)$	t (time)

Why #1: at the peak you know $v_y = 0$, v_{y0} and a_y , and you want t . The height of the peak is not known, so use #1, the one that leaves out Δy .

$$v_y = v_{y0} + a_y t \rightarrow 0 = v_0 \sin \theta - g t_{\text{peak}} \rightarrow t_{\text{peak}} = \frac{v_0 \sin \theta}{g}$$

Step 4 Double it for the whole flight.

The ball lands at the same height it was kicked from, so the trip down takes exactly as long as the trip

$$\text{up: } t_{\text{flight}} = 2t_{\text{peak}} = \frac{2v_0 \sin \theta}{g}$$

Watch out: at the peak $v_y = 0$, but the acceleration is still $a_y = -g$, pointing down. Gravity never switches off. Doubling t_{peak} only works because the launch and landing heights match. And $\Delta y = 0$ over the whole flight means it ends at the height it started, not that the acceleration is zero.

Step 5 Use the flight time in the horizontal column.

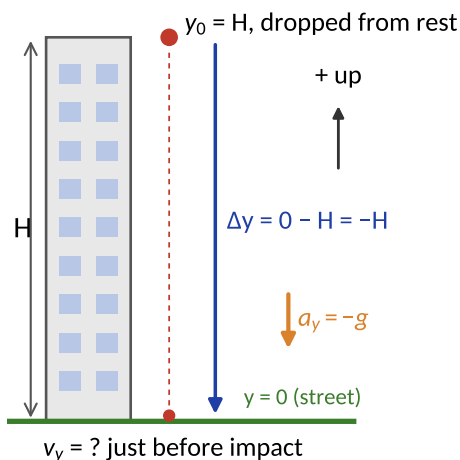
Why #2 (x-form): horizontally you know t and v_{x0} and want Δx ; with $a_x = 0$, #2 becomes $\Delta x = v_{x0}t$.

$$x - x_0 = v_{x0}t + \frac{1}{2}a_x t^2 \rightarrow R = v_{x0}t_{\text{flight}} = (v_0 \cos \theta) \frac{2v_0 \sin \theta}{g} \rightarrow \boxed{R = \frac{2v_0^2 \sin \theta \cos \theta}{g}}$$

Step 6 Now plug in numbers with units.

$$R = \frac{2(22.0 \text{ m/s})^2(\sin 40.0^\circ)(\cos 40.0^\circ)}{(9.80 \text{ m/s}^2)} = \boxed{48.6 \text{ m}}$$

Check: $t_{\text{peak}} = (22.0 \text{ m/s})(\sin 40.0^\circ)/(9.80 \text{ m/s}^2) = 1.443 \text{ s}$, $t_{\text{flight}} = 2.886 \text{ s}$, $v_{x0} = 16.85 \text{ m/s}$, and $(16.85 \text{ m/s})(2.886 \text{ s}) = 48.63 \text{ m} \approx 48.6 \text{ m}$. ✓ (2 sin θ cos θ = sin 2θ , so this also equals $v_0^2 \sin 2\theta / g$.)

4. Dropped from a skyscraper: impact velocity (25 points), variables only

No numbers anywhere. "Physical constants" means g : it is a constant of nature, so it can appear in the answer next to H . You are not given g as a letter; you are expected to know to use it.

No sign convention is given, so **pick one and write it down**. Up is + (the class default).

Origin at the street: the object starts at $y_0 = H$ and ends at $y = 0$, so $y - y_0 = 0 - H = -H$.

Gravity points down, so $a_y = -g$.

Step 1 List the five variables (vertical, so y in place of x).

v_{y0} (initial)	v_y (final)	a_y	t	$y - y_0$
0 (released from rest)	? (wanted)	$-g$	not given, not asked	$0 - H = -H$

Step 2 Pick the equation.

AP sheet	equation	leaves out
#1	$v_y = v_{y0} + a_y t$	$y - y_0$ (Δy)
#2	$y = y_0 + v_{y0} t + \frac{1}{2} a_y t^2$	v_y (final velocity)
#3	$v_y^2 = v_{y0}^2 + 2a_y(y - y_0)$	t (time) ← use this one

Why #3: t is not given and not asked, and you want the final velocity. #3 is the only one without t.

Step 3 Solve with letters.

$$v_y^2 = v_{y0}^2 + 2a_y(y - y_0) = 0^2 + 2(-g)(0 - H) = 2gH \rightarrow v_y = \pm\sqrt{2gH}$$

Step 4 Pick the root and give the direction.

The object is moving **down** just before it hits the street, and with up as +, down is negative. So take the negative root: $v_y = -\sqrt{2gH}$ (downward)

The **speed** is $\sqrt{2gH}$; the minus sign is the direction.

Units check: $\sqrt{(m/s^2)(m)} = \sqrt{m^2/s^2} = m/s$.

Picked down as + instead? Then $a_y = +g$ and $y - y_0 = +H$, and you get $v_y = +\sqrt{2gH}$. That is also right, as long as you wrote down that down is +. A sign only means something once the convention is stated.

Optional extra practice (not scored)**E1. Football: hang time and maximum height**

$$(a) \text{ Hang time} = t_{\text{flight}} = \frac{2v_0 \sin \theta}{g} = \frac{2(22.0 \text{ m/s})(\sin 40.0^\circ)}{(9.80 \text{ m/s}^2)} = \boxed{2.89 \text{ s}}$$

Why #3 (y-form) for (b): at the peak $v_y = 0$, and you want the height without needing t.

$$(b) 0 = (v_0 \sin \theta)^2 - 2g h_{\text{max}} \rightarrow h_{\text{max}} = \frac{v_0^2 \sin^2 \theta}{2g} = \frac{(22.0 \text{ m/s})^2 (\sin 40.0^\circ)^2}{2(9.80 \text{ m/s}^2)} = \boxed{10.2 \text{ m}}$$

E2. Car: stopping time and the doubling question

$$\text{Why \#1: now } t \text{ is wanted and } \Delta x \text{ is not needed. } v_f = v_i + at \rightarrow 0 = v_i + at \rightarrow t = -\frac{v_i}{a} = -\frac{(+27.0 \text{ m/s})}{(-4.50 \text{ m/s}^2)} =$$

$$\boxed{6.00 \text{ s}}$$

$$(b) \Delta x = -\frac{v_i^2}{2a}: v_i \text{ is squared, so doubling it multiplies } \Delta x \text{ by } 2^2 = \boxed{4} \text{ (} 4 \times 81.0 \text{ m} = 324 \text{ m). Not 2.}$$

E3. Skyscraper: velocity as a function of time

$$\text{Why \#1 (y-form): it links } v_y \text{ and } t \text{ directly. Up is +. } v_y = 0 + (-g)t \rightarrow \boxed{v_y(t) = -gt}$$

Why #2 (y-form) for the impact time: you know $y - y_0 = 0 - H = -H$ and want t. $-H = -\frac{1}{2}g t_f^2 \rightarrow t_f = \sqrt{2H/g}$, so it is valid for $\boxed{0 \leq t \leq \sqrt{2H/g}}$. Check: $-g\sqrt{2H/g} = -\sqrt{2gH}$, matching Question 4.

Suggested partial credit (25 points per question, 100 total)

1. Acceleration-time graph

Suggested partial credit	pts
Slope formula $a = \Delta v / \Delta t$ set up with values and units for each of the 4 segments (2 each)	8
Correct slope value with sign and units for each segment (1 each)	4
Four horizontal segments at the correct accelerations (2 each)	8
Changes in acceleration occur at the correct times; no slanted lines	2
Axes labeled with quantity and units; readable scale	3
Total	25

2. Braking car

Suggested partial credit	pts
Starts from $v_f^2 = v_i^2 + 2a\Delta x$ (sheet #3) because time is not involved	5
Sets $v_f = 0$	3
Acceleration negative, with a reason (opposite the motion / slowing down)	4
Solves $\Delta x = -v_i^2 / (2a)$ before substituting	6
Substitution with units in every parenthesis	4
Final answer, 3 significant figures, units, boxed	3
Total	25

3. Football range

Suggested partial credit	pts
Resolves v_0 into $v_0 \cos \theta$ and $v_0 \sin \theta$	4
Time to peak from $v_y = 0$ with sheet #1: $t_{\text{peak}} = v_0 \sin \theta / g$	5
Doubles t_{peak} for the flight time, with the equal-height reason	4
$R = v_{x0}t$ with the expression for t substituted (not a rounded number)	5
Substitution with units in every parenthesis	4
Final answer, 3 significant figures, units, boxed	3
Total	25

4. Skyscraper impact velocity

Suggested partial credit	pts
States a sign convention; $v_{y0} = 0$, a_y and $y - y_0$ set consistently with it (up +: $-g$ and $-H$)	6
Uses sheet #3 (y-form), since t is not involved	4
Algebra to $v_y^2 = 2gH$	5
Root sign matches the stated convention; direction given as downward	6
Final answer in terms of H and g only (g used as the physical constant), boxed	4
Total	25